



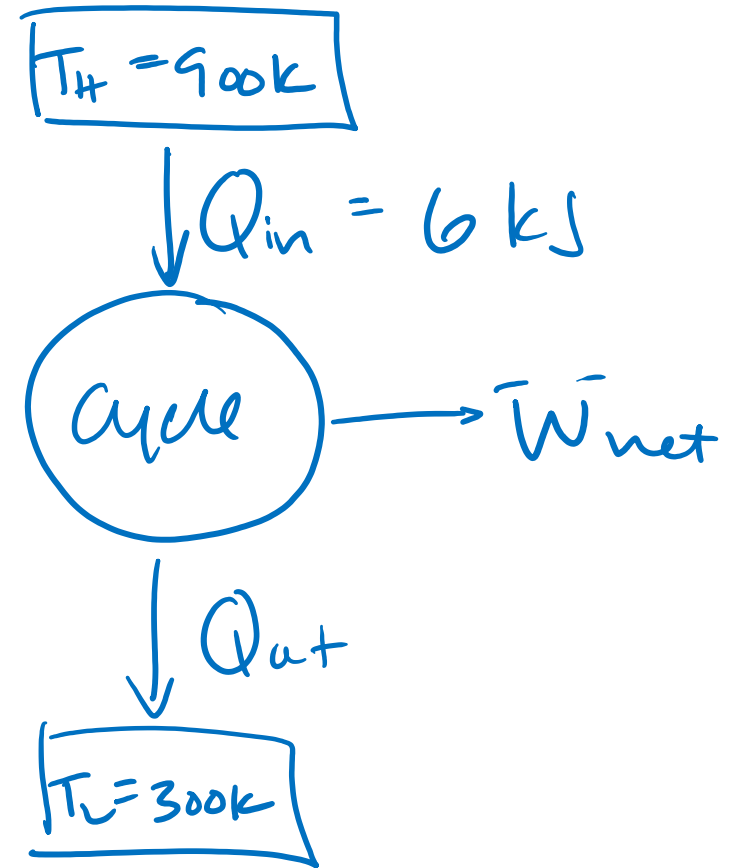
ME 300: Engineering Thermodynamics

Example: Entropy and the Carnot Cycle

Entropy example

- Consider a reversible Carnot cycle:
 - 1-2: reversible, isothermal heat addition
 - 2-3: reversible, adiabatic expansion
 - 3-4: reversible, isothermal heat removal
 - 4-1: reversible, adiabatic compression
- If:
 - $Q_{in} = 6 \text{ kJ}$
 - $T_H = 900 \text{ K}$
 - $T_L = 300 \text{ K}$

- Find:
 - Thermal efficiency $\rightarrow \eta_{th} = \frac{\bar{W}_{net}}{Q_{in}} = 1 - \frac{T_L}{T_H}$
 - $Q_{out} \rightarrow Q_{in} = \bar{W}_{net} + Q_{out}$
 - Change in entropy of the working fluid for each process $\left[\Delta S = \int \frac{\delta Q}{T} \right]_{rev}$



Assume: S.C.S.
: quasi-st.

Entropy example

$$\eta_{th} = 1 - \frac{T_L}{T_H} = 1 - \frac{300}{900} = \frac{2}{3} \rightarrow \eta_{th} = 67\%$$

$$Q_{out} \rightarrow rev.: \quad \frac{Q_L}{Q_H} = \frac{T_L}{T_H} \rightarrow Q_L = Q_H \left(\frac{T_L}{T_H} \right) = 6 \left(\frac{1}{3} \right)$$

$$\underline{\underline{Q_L = 2 \text{ kJ}}}$$

$$\rightarrow Q_{in} = W_{net} + Q_{out} \rightarrow W_{net} = \eta_{th} Q_{in} = \left(\frac{2}{3} \right) (6) = 4 \text{ kJ}$$

$$Q_{out} = Q_{in} - W_{net} = 6 - 4 = \underline{\underline{2 \text{ kJ}}}$$

Entropy example

Process 1-2: $\Delta S' = \int \frac{\delta Q}{T_{rev}} = \frac{6000}{900} = 6.67 \text{ J/K}$

Annotations: Q_{in} points to the numerator, T_H points to the denominator.

2-3: $\Delta S' = 0 \text{ J/K}$ $Q=0$ ✓

3-4: $\Delta S' = \int \frac{\delta Q}{T_{rev}} = \frac{Q_{out}}{T_L} = \frac{-2000}{300} = -6.67 \text{ J/K}$

4-1: $\Delta S' = 0 \text{ J/K}$ ✓

$\sum_{\text{cycle}} \Delta S' = 0$



What is the change in entropy over the entire cycle?

- A. $\Delta S > 0$
- B. $\Delta S < 0$
- C. $\Delta S = 0$
- D. Not enough information